

RAVINDI GUNASEKARA

✉ dananjanagunasekara@gmail.com

📍 Kandy, Sri Lanka

🐙 github.com/ravindidhananjana

☎ +94 74 379 8880

🌐 linkedin.com/in/ravindi-gunasekara-b83483334



👤 SUMMARY

Artificial Intelligence and Machine Learning undergraduate with interests in computer vision, deep learning, multimodal machine learning, and generative AI. Experienced in developing end-to-end ML solutions, computer vision applications, and AI-driven systems using Python and modern AI frameworks. Passionate about applying AI to real-world challenges in healthcare, remote sensing, space technology, and autonomous systems through applied research and engineering.

🧰 EXPERIENCE

07/2026 – Present
Remote

FlyRank AI

Machine Learning Intern

- Researching and developing ML approaches for **SEO ranking optimization**, exploring how learned models can outperform traditional hand-crafted ranking rules.
- Supporting ML workflows including **data preparation, feature engineering, model training, and evaluation** for ranking-related tasks.
- Analyzing ranking signals and experimenting with data-driven approaches to improve automated search ranking decisions.
- Evaluating machine learning models against rule-based approaches to understand performance improvements in ranking systems.

🧠 TECHNICAL SKILLS

Programming

- Python
- SQL

Databases

- MongoDB
- SQL

Machine Learning

- Deep Learning
- Model Development
- Feature Engineering
- Model Evaluation

Computer Vision

- Image Processing
- OpenCV
- CNN Architectures

Frameworks & Tools

- TensorFlow
- PyTorch
- Keras
- Scikit-learn
- XGBoost
- Pandas
- NumPy
- Streamlit
- Git

Generative AI

- Large Language Models
- Google Gemini API
- Retrieval-Augmented Generation (RAG)

Research Areas

- Remote Sensing
- Earth Observation
- Autonomous Systems
- Space Tech
- Medical AI

PROJECTS & RESEARCH EXPERIENCE

01/2026 – 05/2026

Acute Myeloid Leukemia Classification Using Multimodal Machine Learning

Machine Learning & Medical AI Research Project

- Developed a multimodal ML system combining ~81,000 blood smear microscopy images (**TCIA dataset**) with clinical patient data for AML subtype classification.
- Extracted deep image features using pretrained CNNs (**EfficientNetBo, ResNet50, MobileNetV2**) and fused them with engineered clinical features.
- Evaluated **Random Forest, XGBoost, and MLP classifiers** using stratified cross validation, achieving 86.8% test accuracy with multimodal fusion outperforming single-modality approaches.
- Deployed a Streamlit prediction application with an integrated AI assistant for interactive classification

04/2026 – 04/2026

Lunar Landing Site Risk Analysis Using Computer Vision

Computer Vision & Image Processing Project

- Developed a computer vision system for lunar landing site risk assessment using real **NASA/LROC** surface imagery.
- Applied image processing techniques including **CLAHE enhancement, edge detection, and shadow/roughness analysis** to identify hazardous terrain features such as craters and uneven surfaces.
- Generated terrain risk maps by combining visual features to identify safer and higher risk landing regions. Explored applications in autonomous lunar landing and planetary surface analysis

06/2026 – 07/2026

Sri Lankan Final MBBS Clinical Case Simulator

Generative AI Healthcare Application

- Built a generative AI clinical case simulator using Google Gemini to dynamically role play both patient and examiner interactions for Sri Lankan Final MBBS exam preparation.
- Designed syllabus-aligned case generation across Medicine, Surgery, Pediatrics, Obstetrics, and Gynecology with hidden diagnoses to simulate real examination scenarios.
- Implemented multilingual voice-based interaction (Sinhala, Singlish, English) using the **Web Speech API** for hands-free clinical conversations.
- Deployed the application using Streamlit for real-world educational use.

EDUCATION | CERTIFICATIONS

08/2024 – 09/2027

BSc (Honours) in Computer Science with Artificial Intelligence

Coventry University, United Kingdom (Delivered at NIBM Sri Lanka)

Expected: SEP 2027

Academic Pathway:

Diploma in Computer Science with AI (GPA: 3.89)

Higher National Diploma in Computer Science with AI (GPA: 3.87)

2024 – 2024

Short Course in Information and Communication Technology

Open University, Kandy, Sri Lanka

2014 – 2023

Girls High School , Kandy

Completed GCE O/L and GCE A/L successfully (2014-2023)